

AD23532 – Principles of Data Science**EX:1:Title: Data Preprocessing and Cleaning****Learning Objectives :**

The objective of this experiment is to understand and perform the essential steps involved in data preprocessing. The experiment provides a foundation for preparing high-quality datasets by handling missing values, removing duplicate records, converting data types, encoding categorical variables, scaling numerical features, and visualizing data before applying machine learning algorithms.

Methods Used / Python Libraries Involved and Purpose :

NumPy is used for numerical computations and mathematical operations. It provides efficient array processing and supports statistical calculations, making it useful for handling numerical data during preprocessing.

Pandas is used to load the dataset into a DataFrame and perform various data manipulation tasks such as viewing records, handling missing values, removing duplicate entries, and converting data types. It is one of the most important libraries for working with structured data.

Matplotlib.pyplot is used to create basic visualizations such as histograms and other graphs. These visualizations help in understanding the distribution of numerical features and identifying patterns in the dataset.

Seaborn is a statistical data visualization library built on top of Matplotlib. It is used to create attractive and informative visualizations such as count plots, box plots, and correlation heatmaps, which help in analyzing relationships between variables.

The **StandardScaler** class from the Scikit-learn library is used to standardize numerical features such as *Tenure*, *MonthlyCharges*, and *TotalCharges*. Standardization transforms the data to have a mean of zero and a standard deviation of one, which improves the performance of many machine learning algorithms.

LabelEncoder is used to convert binary categorical variables such as **Gender**, **Partner**, **Dependents**, and **Churn** into numerical values. Since machine learning models cannot process text directly, encoding converts categorical data into a numerical format.

One-Hot Encoding, implemented using the **get_dummies()** function in Pandas, is applied to categorical variables with multiple categories, such as **Contract**, **PaymentMethod**, and **InternetService**. This method creates separate binary columns for each category, allowing machine learning algorithms to interpret categorical information correctly.

To handle missing data, the **Median** method is used for numerical columns because it is less affected by extreme values or outliers. The **Mode** method is used for categorical columns because it replaces missing values with the most frequently occurring category, preserving the characteristics of the dataset.

Finally, a **Correlation Heatmap** is generated using Seaborn to visualize the relationships among numerical features. It helps identify positive and negative correlations between variables, providing useful insights for feature selection and data analysis before building machine learning models.

PROGRAM CODE :

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.preprocessing import LabelEncoder, StandardScaler

df = pd.read_csv("Customer_Churn.csv")

print("First 10 Records")

print(df.head(10))

print("\nLast 10 Records")

print(df.tail(10))

print("\nShape of Dataset")

print(df.shape)

print("\nColumn Names")

print(df.columns)

print("\nDataset Information")

print(df.info())

print("\nSummary Statistics")

print(df.describe())

print("\nMissing Values")

print(df.isnull().sum())

print("\nDuplicate Records")

print(df.duplicated().sum())

df = df.drop_duplicates()

df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")

numerical_columns = df.select_dtypes(include=["int64", "float64"]).columns

categorical_columns = df.select_dtypes(include=["object"]).columns
```

```
for col in numerical_columns:

    df[col] = df[col].fillna(df[col].median())

for col in categorical_columns:

    df[col] = df[col].fillna(df[col].mode()[0])

print("\nMissing Values After Cleaning")

print(df.isnull().sum())

label_columns = ["gender", "Partner", "Dependents", "Churn"]

le = LabelEncoder()

for col in label_columns:

    if col in df.columns:

        df[col] = le.fit_transform(df[col])

onehot_columns = ["Contract", "PaymentMethod", "InternetService"]

existing_columns = [col for col in onehot_columns if col in df.columns]

df = pd.get_dummies(df, columns=existing_columns, drop_first=True)

scaler = StandardScaler()

scale_columns = ["tenure", "MonthlyCharges", "TotalCharges"]

existing_scale = [col for col in scale_columns if col in df.columns]

df[existing_scale] = scaler.fit_transform(df[existing_scale])

plt.figure(figsize=(6,4))

sns.histplot(df["MonthlyCharges"], kde=True)

plt.title("Distribution of MonthlyCharges")

plt.show()

plt.figure(figsize=(6,4))

sns.histplot(df["tenure"], kde=True)

plt.title("Distribution of Tenure")

plt.show()

if "Contract" in pd.read_csv("Customer_Churn.csv").columns:
```

```
plt.figure(figsize=(8,5))

sns.countplot(x="Contract", hue="Churn", data=pd.read_csv("Customer_Churn.csv"))

plt.title("Contract vs Churn")

plt.xticks(rotation=20)

plt.show()

plt.figure(figsize=(6,4))

sns.boxplot(x="Churn", y="MonthlyCharges", data=df)

plt.title("MonthlyCharges by Churn")

plt.show()

plt.figure(figsize=(12,8))

sns.heatmap(df.select_dtypes(include=np.number).corr(), annot=True, cmap="coolwarm")

plt.title("Correlation Heatmap")

plt.show()

print("\nFinal Shape:", df.shape)

df.to_csv("Cleaned_Customer_Churn.csv", index=False)

print("\nCleaned dataset saved as 'Cleaned_Customer_Churn.csv'")
```

OUTPUT :

```
***** FIRST 10 RECORDS *****

customerID  gender  SeniorCitizen  Partner  Dependents  tenure  PhoneService  \
0  7598-VWEG  Female           0      Yes         No         1         No
1  5575-QWDE  Male             0      No          No         34         Yes
2  3668-QPVBK Male             0      No          No         2         Yes
3  7795-CFOCN Male             0      No          No         45         No
4  9237-HQITU Female           0      No          No         2         Yes
5  9585-CISXC Female           0      No          No         8         Yes
6  2452-KIDWK Male             0      No          Yes        22         Yes
7  6713-CXQKP Female           0      No          No         10         No
8  7892-PDOOP Female           0      Yes         No         28         Yes
9  6388-TABGU Male             0      No          Yes        62         Yes

MultipleLines  InternetService  OnlineSecurity  ...  DeviceProtection  \
0  No phone service  DSL          No          ...          No
1  No                DSL          Yes          ...          Yes
2  No                DSL          Yes          ...          No
3  No phone service  DSL          Yes          ...          Yes
4  No                Fiber optic No          ...          No
5  Yes               Fiber optic No          ...          Yes
6  Yes               Fiber optic No          ...          No
7  No phone service  DSL          Yes          ...          No
8  Yes               Fiber optic No          ...          Yes
9  No                DSL          Yes          ...          No

TechSupport  StreamingTV  StreamingMovies  Contract  PaperlessBilling  \
0  No          No          No  Month-to-month  Yes
1  No          No          No  One year       No
2  No          No          No  Month-to-month  Yes
3  Yes         No          No  One year       No
4  No          No          No  Month-to-month  Yes
5  No          Yes         Yes  Month-to-month  Yes
6  No          Yes         No  Month-to-month  Yes
7  No          No          No  Month-to-month  No
```

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes
5	Electronic check	99.65	820.5	Yes
6	Credit card (automatic)	89.10	1949.4	No
7	Mailed check	29.75	301.9	No
8	Electronic check	104.00	3046.05	Yes
9	Bank transfer (automatic)	56.15	3487.95	No

[10 rows x 21 columns]

***** LAST 10 RECORDS *****

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure
7033	9767-FFLEM	Male	0	No	No	38
7034	0639-TSIQW	Female	0	No	No	67
7035	8456-QDAVC	Male	0	No	No	19
7036	7750-EYXWZ	Female	0	No	No	12
7037	2569-WGERO	Female	0	No	No	72
7038	6840-RESVB	Male	0	Yes	Yes	24
7039	2234-XADUH	Female	0	Yes	Yes	72
7040	4801-JZAZL	Female	0	Yes	Yes	11
7041	8361-LTHKD	Male	1	Yes	No	4
7042	3186-AJIEK	Male	0	No	No	66

	PhoneService	MultipleLines	InternetService	OnlineSecurity	...
7033	Yes	No	Fiber optic	No	...
7034	Yes	Yes	Fiber optic	Yes	...
7035	Yes	No	Fiber optic	No	...
7036	No	No phone service	DSL	No	...
7037	Yes	No	No	No internet service	...
7038	Yes	Yes	DSL	Yes	...
7039	Yes	Yes	Fiber optic	No	...
7040	No	No phone service	DSL	Yes	...

	DeviceProtection	TechSupport	StreamingTV
7833	No	No	No
7834	Yes	No	Yes
7835	No	No	Yes
7836	Yes	Yes	Yes
7837	No internet service	No internet service	No internet service
7838	Yes	Yes	Yes
7839	Yes	No	Yes
7840	No	No	No
7841	No	No	No
7842	Yes	Yes	Yes

	StreamingMovies	Contract	PaperlessBilling
7833	No	Month-to-month	Yes
7834	No	Month-to-month	Yes
7835	No	Month-to-month	Yes
7836	Yes	One year	No
7837	No internet service	Two year	Yes
7838	Yes	One year	Yes
7839	Yes	One year	Yes
7840	No	Month-to-month	Yes
7841	No	Month-to-month	Yes
7842	Yes	Two year	Yes

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
7833	Credit card (automatic)	69.50	2625.25	No
7834	Credit card (automatic)	102.95	6886.25	Yes
7835	Bank transfer (automatic)	78.70	1495.1	No
7836	Electronic check	60.65	743.3	No
7837	Bank transfer (automatic)	21.15	1419.4	No
7838	Mailed check	84.00	1990.5	No
7839	Credit card (automatic)	103.20	7362.9	No
7840	Electronic check	29.00	346.45	No
7841	Mailed check	74.40	306.6	Yes
7842	Bank transfer (automatic)	105.65	6044.5	No

[10 rows x 21 columns]

***** SUMMARY STATISTICS *****

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure
count	7843	7843	7843.000000	7843	7843	7843.000000
unique	7843	2	NaN	2	2	NaN
top	3186-AJIEK	Male	NaN	No	No	NaN
freq	1	3555	NaN	3641	4933	NaN
mean	NaN	NaN	0.162147	NaN	NaN	32.371149
std	NaN	NaN	0.368612	NaN	NaN	24.550481
min	NaN	NaN	0.000000	NaN	NaN	0.000000
25%	NaN	NaN	0.000000	NaN	NaN	9.000000
50%	NaN	NaN	0.000000	NaN	NaN	29.000000
75%	NaN	NaN	0.000000	NaN	NaN	55.000000
max	NaN	NaN	1.000000	NaN	NaN	72.000000

	PhoneService	MultipleLines	InternetService	OnlineSecurity	...
count	7843	7843	7843	7843	...
unique	2	3	3	3	...
top	Yes	No	Fiber optic	No	...
freq	6361	3390	3096	3498	...
mean	NaN	NaN	NaN	NaN	...
std	NaN	NaN	NaN	NaN	...
min	NaN	NaN	NaN	NaN	...
25%	NaN	NaN	NaN	NaN	...
50%	NaN	NaN	NaN	NaN	...
75%	NaN	NaN	NaN	NaN	...
max	NaN	NaN	NaN	NaN	...

	DeviceProtection	TechSupport	StreamingTV	StreamingMovies
count	7843	7843	7843	7843
unique	3	3	3	3
top	No	No	No	No
freq	3095	3473	2810	2785
mean	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN

***** MISSING VALUES *****

customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	0
Churn	0
dtype: int64	

***** DUPLICATE RECORDS *****

0
Duplicates after removal: 0

***** MISSING VALUES AFTER CLEANING *****

customerID	0
gender	0
SeniorCitizen	0

[10 rows x 21 columns]

[10 rows x 21 columns]

***** SHAPE *****

(7843, 21)

***** COLUMN NAMES *****

['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'MultipleLines',

***** DATASET INFORMATION *****

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 7843 entries, 0 to 7842

Data columns (total 21 columns):

#	Column	Non-Null Count	Dtype
0	customerID	7843 non-null	object
1	gender	7843 non-null	object
2	SeniorCitizen	7843 non-null	int64
3	Partner	7843 non-null	object
4	Dependents	7843 non-null	object
5	tenure	7843 non-null	int64
6	PhoneService	7843 non-null	object
7	MultipleLines	7843 non-null	object
8	InternetService	7843 non-null	object
9	OnlineSecurity	7843 non-null	object
10	OnlineBackup	7843 non-null	object
11	DeviceProtection	7843 non-null	object
12	TechSupport	7843 non-null	object
13	StreamingTV	7843 non-null	object
14	StreamingMovies	7843 non-null	object
15	Contract	7843 non-null	object
16	PaperlessBilling	7843 non-null	object
17	PaymentMethod	7843 non-null	object
18	MonthlyCharges	7843 non-null	float64

	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges
count	7843	7843	7843	7843.000000
unique	3	2	4	NaN
top	Month-to-month	Yes	Electronic check	NaN
freq	3875	4171	2365	NaN
mean	NaN	NaN	NaN	64.761692
std	NaN	NaN	NaN	30.090047
min	NaN	NaN	NaN	18.250000
25%	NaN	NaN	NaN	35.500000
50%	NaN	NaN	NaN	70.350000
75%	NaN	NaN	NaN	89.850000
max	NaN	NaN	NaN	118.750000

TotalCharges Churn

	TotalCharges	Churn
count	7843	7843
unique	6531	2
top	No	No
freq	11	5174
mean	NaN	NaN
std	NaN	NaN
min	NaN	NaN
25%	NaN	NaN
50%	NaN	NaN
75%	NaN	NaN
max	NaN	NaN

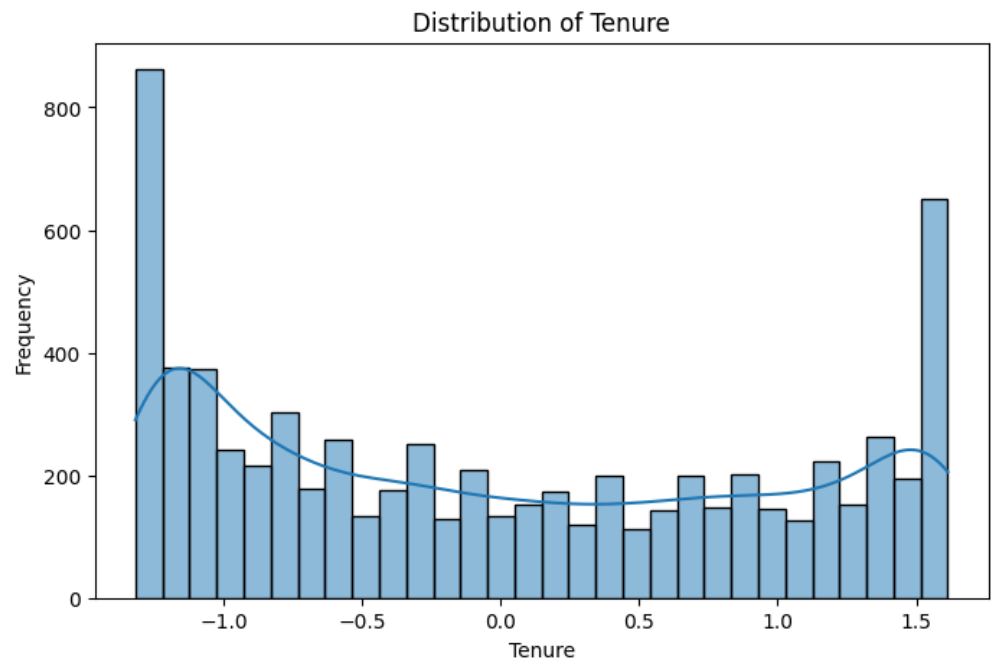
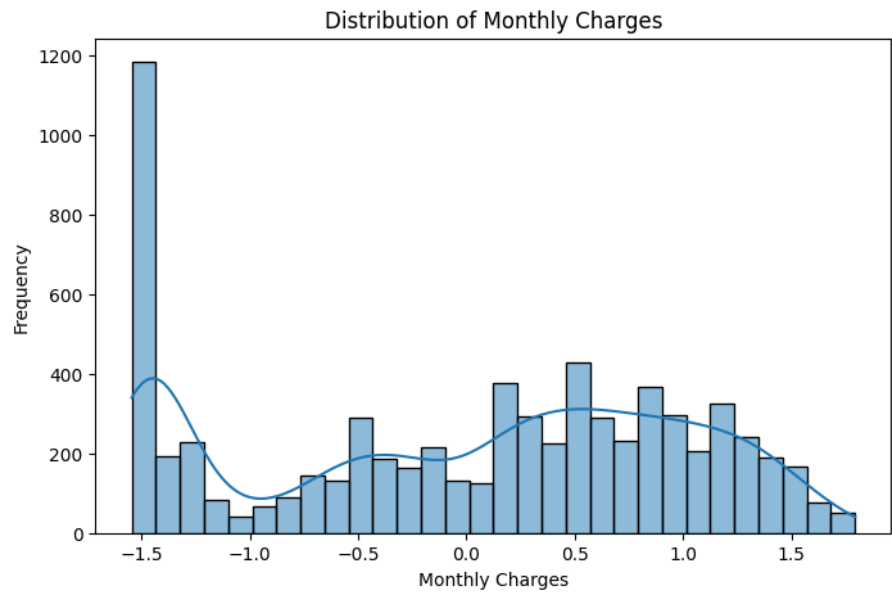
[11 rows x 21 columns]

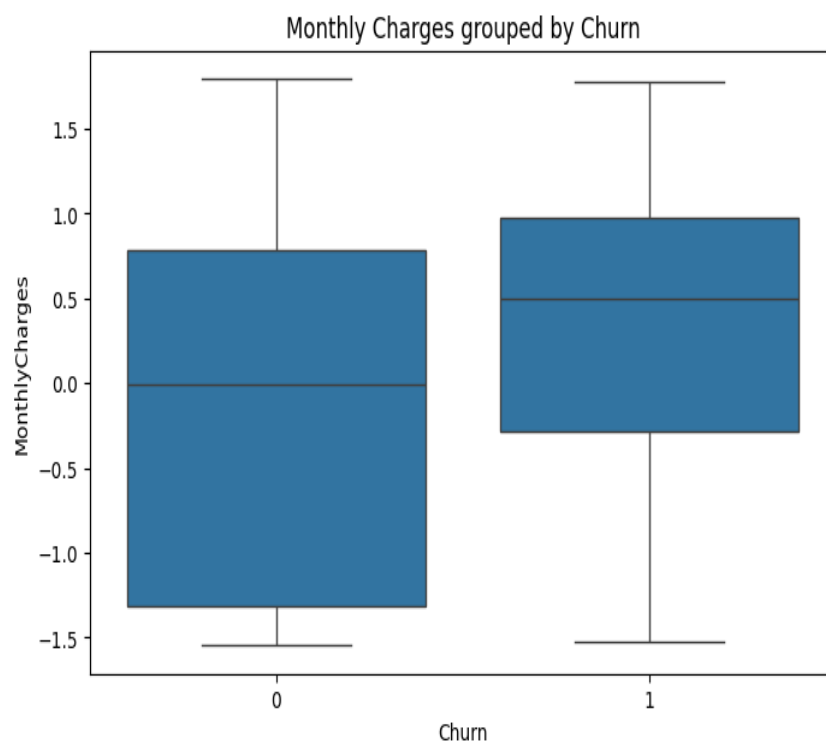
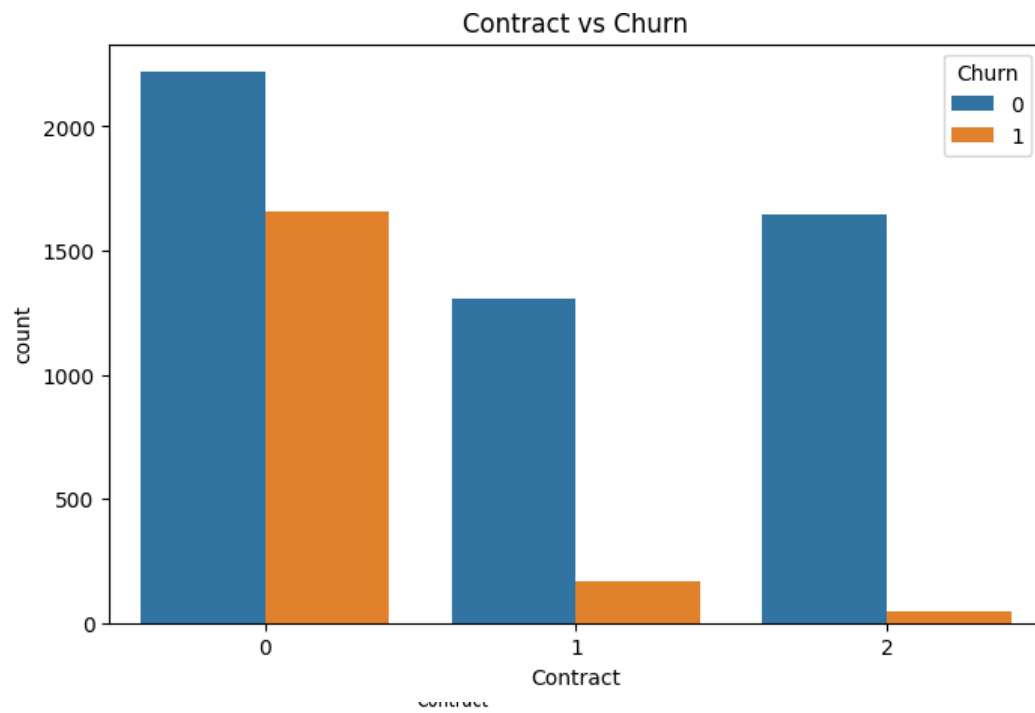
***** MISSING VALUES *****

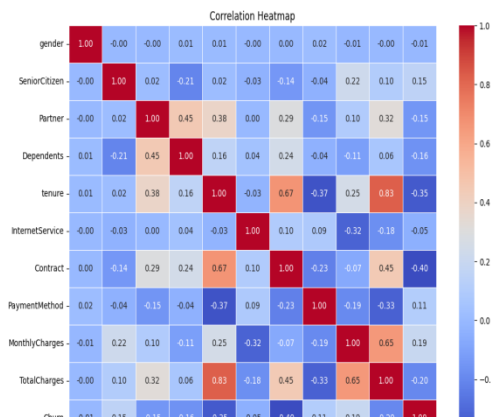
customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0

```
... ----- MISSING VALUES AFTER CLEANING -----
customerID      0
gender           0
SeniorCitizen   0
Partner         0
Dependents      0
Tenure          0
PhoneService    0
MultipleLines   0
InternetService 0
OnlineSecurity  0
OnlineBackup    0
DeviceProtection 0
TechSupport     0
StreamingTV     0
StreamingMovies 0
Contract        0
PaperlessBilling 0
PaymentMethod   0
MonthlyCharges  0
TotalCharges    0
Churn           0
dtype: int64

Categorical Encoding Completed.
Standardization Completed.
```







===== FINAL CLEANED DATASET =====

```

customerID  gender  SeniorCitizen  Partner  Dependents  tenure  \
0  7590-VHVEG      0              0        1          0 -1.277445
1  5575-GNVDE      1              0        0          0  0.066327
2  3668-QPYBK      1              0        0          0 -1.236724
3  7795-CFOCW      1              0        0          0  0.514251
4  9237-HQITU      0              0        0          0 -1.236724

PhoneService  MultipleLines  InternetService  OnlineSecurity  ...  \
0  No  No phone service      0          No  ...
1  Yes  No                  0          Yes  ...
2  Yes  No                  0          Yes  ...
3  No  No phone service      0          Yes  ...
4  Yes  No                  1          No  ...

DeviceProtection  TechSupport  StreamingTV  StreamingMovies  Contract  \
0  No  No  No  No  No  0
1  Yes  No  No  No  No  1
2  No  No  No  No  No  0
3  Yes  Yes  No  No  No  1
4  No  No  No  No  No  0

PaperlessBilling  PaymentMethod  MonthlyCharges  TotalCharges  Churn
0  Yes  2  -1.160323  -0.994242  0
1  No  3  -0.259629  -0.173244  0
2  Yes  3  -0.362660  -0.959674  1
3  No  0  -0.746535  -0.194766  0
4  Yes  2  0.197365  -0.940470  1

[5 rows x 21 columns]

Final Shape: (7043, 21)

Cleaned dataset saved as 'Cleaned_Customer_Churn.csv'

```

Learning Outcome

From this experiment, I learned the importance of data preprocessing in the data science workflow. I gained practical knowledge in loading datasets, identifying and handling missing values, removing duplicate records, converting data types, encoding categorical variables, scaling numerical features, and creating meaningful visualizations. These preprocessing techniques improve the quality of the dataset and make it suitable for machine learning and predictive analytics.